Notes on Varieties of Semiring Solvers

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Contents

1 Preliminaries

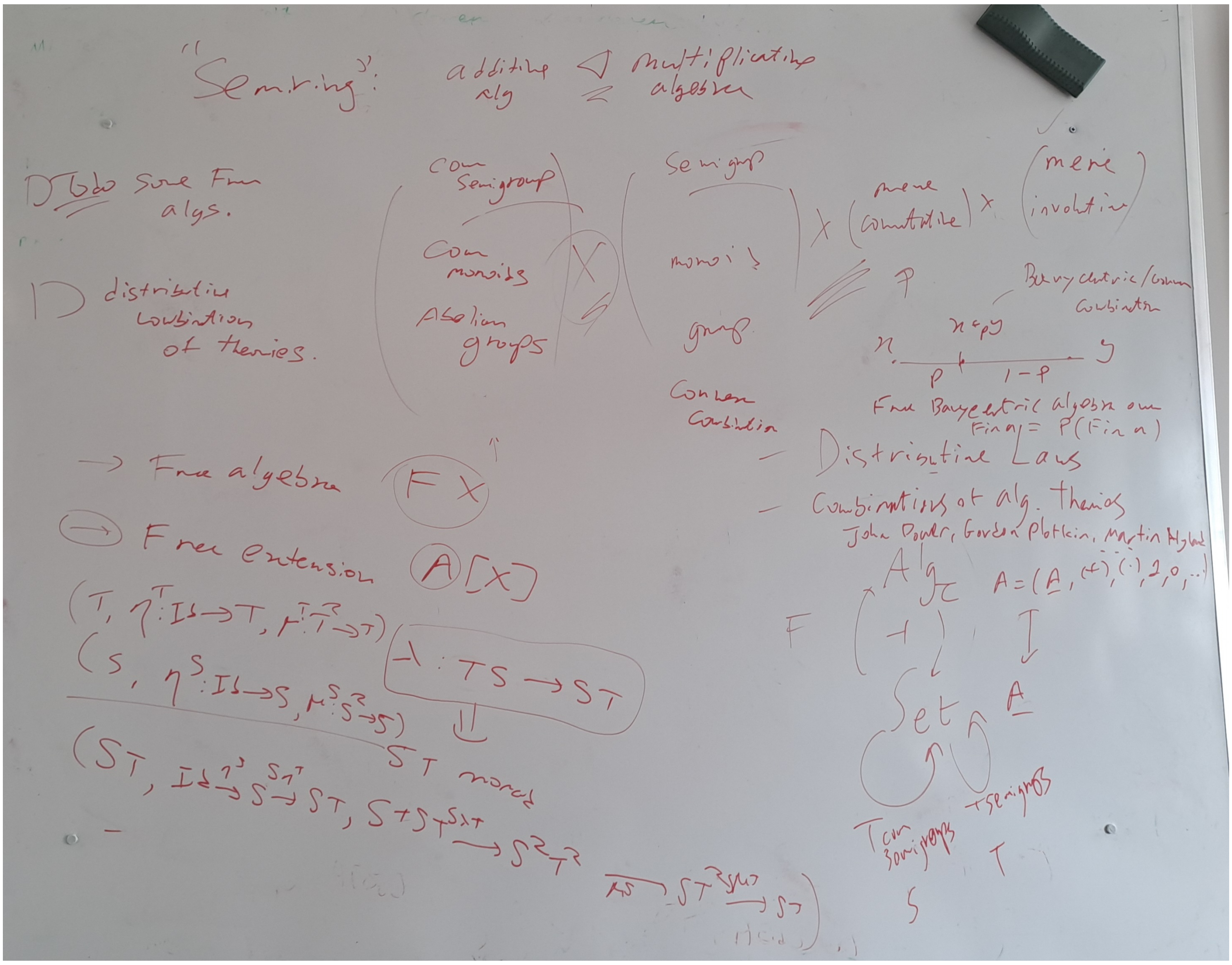

Figure 1: Blackboard at the start of the internship.

For convenience, we recall some definitions we'll be using.

1.1 Algebraic Structures

Definition 1 (Semigroup)

A *semigroup* is a set S equipped with an associative binary operation $\cdot : S \times S \to S$ such that for all $a, b, c \in S$:

- $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ (associativity)

Definition 2 (Monoid)

A *monoid* is a set M equipped with an associative binary operation $\cdot : M \times M \to M$ and an identity element $e \in M$ such that for all $a, b, c \in M$:

- $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ (associativity)
- $e \cdot a = a \cdot e = a$ (identity element)

Definition 3 (Group)

A *group* is a set G equipped with an associative binary operation $\cdot : G \times G \to G$, an identity element $e \in G$, and an inverse operation such that for all $a, b, c \in G$:

- $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ (associativity)
- $e \cdot a = a \cdot e = a$ (identity element)
- For each $a \in G$, there exists an element $a^{-1} \in G$ such that $a \cdot a^{-1} = a^{-1} \cdot a = e$ (inverse element)

A group is called *abelian* if its operation is commutative, i.e., for all $a, b \in G$, $a \cdot b = b \cdot a$.

Definition 4 (Semiring)

A *semiring* is a set R equipped with two binary operations: addition $+$ and multiplication $\cdot$, such that:

- $(R, +)$ is a commutative monoid with identity element 0.
- $(R, \cdot)$ is a monoid with identity element 1.
- Multiplication distributes over addition from both sides: for all $a, b, c \in R$,

$$a \cdot (b + c) = a \cdot b + a \cdot c,$$
$$(a + b) \cdot c = a \cdot c + b \cdot c.$$

- The additive identity 0 is an absorbing element for multiplication: for all $a \in R$, $0 \cdot a = a \cdot 0 = 0$.

Definition 5 (Operad)

An *operad* $\mathcal{O}$ consists of:

- A collection of sets $\{\mathcal{O}_n\}_{n \geq 0}$, where $\mathcal{O}_n$ is the set of n-ary operations,
- A composition function $\gamma : \mathcal{O}_n \times \mathcal{O}_{k_1} \times \cdots \times \mathcal{O}_{k_n} \to \mathcal{O}_{k_1 + \cdots + k_n}$,
- An identity element $e \in \mathcal{O}_1$,

satisfying the following axioms:

- Associativity: For all $f \in \mathcal{O}_n$, $g_i \in \mathcal{O}_{k_i}$, and $h_{ij} \in \mathcal{O}_{m_{ij}}$, we have

$$\gamma(f; g_1, \ldots, g_n) = f(g_1(h_{11}, \ldots, h_{1k_1}), \ldots, g_n(h_{n1}, \ldots, h_{nk_n})).$$

- Identity: For all $f \in \mathcal{O}_n$, we have

$$f(e, \ldots, e) = f.$$

Definition 6 (Algrebraic Theory)

An *algebraic theory* $\mathcal{T}$ consists of:

- A collection of sets $\{\mathcal{T}_n\}_{n \geq 0}$, where $\mathcal{T}_n$ is the set of n-ary operations,
- A collection of equations between terms built from these operations,

such that the operations and equations define a class of algebras (models) that satisfy the given equations.

Example (Operad from an Algebraic Theory)

Given an algebraic theory $\mathcal{T}$, we can construct an operad $\mathcal{O}^{\mathcal{T}}$ where:

- $\mathcal{O}^{\mathcal{T}}_n$ consists of the n-ary operations in $\mathcal{T}$,
- The composition function γ is defined by substituting operations according to the equations of the theory,
- The identity element e corresponds to the identity operation in the theory.

1.2 Categories

Definition 7 (Category)

A *category* $\mathcal{C}$ consists of:

- A class of objects $\mathrm{Ob}(\mathcal{C})$,
- For each pair of objects $a, b \in \mathrm{Ob}(\mathcal{C})$, a set of morphisms $\mathrm{Hom}_{\mathcal{C}}(a, b)$,
- For each object $a \in \mathrm{Ob}(\mathcal{C})$, an identity morphism $\mathrm{id}_a \in \mathrm{Hom}_{\mathcal{C}}(a, a)$,
- A composition operation $\circ$ such that for morphisms $f : a \to b$ and $g : b \to c$, there is a morphism $g \circ f : a \to c$,

satisfying the following axioms:

- Associativity: For morphisms $f : a \to b$, $g : b \to c$, and $h : c \to d$, we have $h \circ (g \circ f) = (h \circ g) \circ f$.
- Identity: For any morphism $f : a \to b$, we have $\mathrm{id}_b \circ f = f = f \circ \mathrm{id}_a$.

Example (Set Category)

The category **Set** has sets as objects and functions as morphisms. The identity morphism for a set a is the identity function $\mathrm{id}_a : a \to a$ defined by $\mathrm{id}_a(x) = x$ for all $x \in a$. Composition of morphisms is given by function composition.

Definition 8 (Product)

Given two objects a and b in a category $\mathcal{C}$, a *product* of a and b is an object $a \times b$ together with two morphisms $\pi_a : a \times b \to a$ and $\pi_b : a \times b \to b$ such that for any object c and morphisms $f : c \to a$ and $g : c \to b$, there exists a unique morphism $\langle f, g \rangle : c \to a \times b$ making the following diagram commute:

$$\begin{tikzcd} & c \arrow[dl, "f"'] \arrow[d, dashed, "\langle f,g\rangle"] \arrow[dr, "g"] & \\ a & a \times b \arrow[l, "\pi_a"] \arrow[r, "\pi_b"'] & b \end{tikzcd}$$

Definition 9 (Coproduct)

Given two objects a and b in a category $\mathcal{C}$, a *coproduct* of a and b is an object $a+b$ together with two morphisms $\iota_a : a \to a+b$ and $\iota_b : b \to a+b$ such that for any object c and morphisms $f : a \to c$ and $g : b \to c$, there exists a unique morphism $[f,g] : a+b \to c$ making the following diagram commute:

$$\begin{tikzcd} a \arrow[r, "\iota_a"] \arrow[dr, "f"'] & a+b \arrow[d, dashed, "{[f,g]}"] & b \arrow[l, "\iota_b"'] \arrow[dl, "g"] \\ & c & \end{tikzcd}$$

Definition 10 (Monoidal Category)

A *monoidal category* is a category $\mathcal{C}$ equipped with a bifunctor $\otimes : \mathcal{C} \times \mathcal{C} \to \mathcal{C}$, an object I called the unit object, and natural isomorphisms (called associators and unitors) that satisfy certain coherence conditions. Specifically, for all objects a, b, c in $\mathcal{C}$, there are natural isomorphisms:

- Associator: $\alpha_{a,b,c} : (a \otimes b) \otimes c \to a \otimes (b \otimes c)$,
- Left unitor: $\lambda_a : I \otimes a \to a$,
- Right unitor: $\rho_a : a \otimes I \to a$,

such that the following diagrams commute:

$$\begin{tikzcd} ((a \otimes b) \otimes c) \otimes d \arrow[r, "\alpha_{a,b,c} \otimes 1_d"] & (a \otimes (b \otimes c)) \otimes d \arrow[r, "\alpha_{a,b \otimes c,d}"] & a \otimes ((b \otimes c) \otimes d) \\ (a \otimes b) \otimes (c \otimes d) \arrow[u, "\alpha_{a,b,c \otimes d}"] & & a \otimes (b \otimes (c \otimes d)) \arrow[u, "1_a \otimes \alpha_{b,c,d}"'] \end{tikzcd}$$

$$\begin{tikzcd} (I \otimes a) \otimes b \arrow[r, "\alpha_{I,a,b}"] & I \otimes (a \otimes b) & a = a \otimes I \arrow[l, "\rho_a"'] \arrow[r, "\lambda_a"] & a \otimes I \arrow[r, "\alpha_{a,I,I}"] & (a \otimes I) \otimes I \end{tikzcd}$$

Definition 11 (Functor)

Given two categories $\mathcal{C}$ and $\mathcal{D}$, a *functor* $F : \mathcal{C} \to \mathcal{D}$ consists of:

- A mapping $F : \mathrm{Ob}(\mathcal{C}) \to \mathrm{Ob}(\mathcal{D})$ that assigns to each object a in $\mathcal{C}$ an object $F(a)$ in $\mathcal{D}$,
- For each pair of objects $a, b \in \mathrm{Ob}(\mathcal{C})$, a mapping $F : \mathrm{Hom}_{\mathcal{C}}(a,b) \to \mathrm{Hom}_{\mathcal{D}}(F(a), F(b))$ that assigns to each morphism $f : a \to b$ in $\mathcal{C}$ a morphism $F(f) : F(a) \to F(b)$ in $\mathcal{D}$,

such that for all objects a, b, c in $\mathcal{C}$ and morphisms $f : a \to b$, $g : b \to c$, the following conditions hold:

- $F(g \circ f) = F(g) \circ F(f)$ (preservation of composition),
- $F(\mathrm{id}_a) = \mathrm{id}_{F(a)}$ (preservation of identity).

For a category $\mathcal{C}$, the *identity functor* $\mathrm{Id}_{\mathcal{C}} : \mathcal{C} \to \mathcal{C}$ is defined by $\mathrm{Id}_{\mathcal{C}}(a) = a$ for all objects a and $\mathrm{Id}_{\mathcal{C}}(f) = f$ for all morphisms f.

- A functor $F : \mathcal{C} \to \mathcal{D}$ is *faithful* if, for every pair of objects $a, b \in \mathrm{Ob}(\mathcal{C})$, the map $F : \mathrm{Hom}_{\mathcal{C}}(a,b) \to \mathrm{Hom}_{\mathcal{D}}(Fa, Fb)$ is injective.
- A functor $F : \mathcal{C} \to \mathcal{D}$ is *full* if, for every pair of objects $a, b \in \mathrm{Ob}(\mathcal{C})$, the map $F : \mathrm{Hom}_{\mathcal{C}}(a,b) \to \mathrm{Hom}_{\mathcal{D}}(Fa, Fb)$ is surjective.
- A functor that is both full and faithful is called *fully faithful*.

Notation: We often denote the image of an object a under a functor F as Fa instead of $F(a)$ and the image of a morphism $f : a \to b$ as $Ff : Fa \to Fb$.

Definition 12 (Natural Transformation)

Given two functors $F, G : \mathcal{C} \to \mathcal{D}$ between categories $\mathcal{C}$ and $\mathcal{D}$, a *natural transformation* $\eta : F \to G$ consists of a family of morphisms $\{\eta_a : Fa \to Ga\}_{a \in \mathrm{Ob}(\mathcal{C})}$ such that for every morphism $f : a \to b$ in $\mathcal{C}$, the following diagram commutes:

$$\begin{tikzcd} Fa \arrow[r, "Ff"] \arrow[d, "\eta_a"'] & Fb \arrow[d, "\eta_b"] \\ Ga \arrow[r, "Gf"'] & Gb \end{tikzcd}$$

Definition 13 (Monad)

A *monad* on a category $\mathcal{C}$ is a triple (T, η, μ) where:

- $T : \mathcal{C} \to \mathcal{C}$ is a functor,
- $\eta : \mathrm{Id}_{\mathcal{C}} \to T$ is a natural transformation called the unit,
- $\mu : T^2 \to T$ is a natural transformation called the multiplication,

such that the following diagrams commute:

$$\begin{tikzcd} T \arrow[r, "T\eta"] \arrow[d, "\eta T"] & T^2 \arrow[d, "\mu"] \\ T^2 \arrow[r, "\mu"] & T \end{tikzcd} \qquad \begin{tikzcd} T^3 \arrow[r, "T\mu"] \arrow[d, "\mu T"] & T^2 \arrow[d, "\mu"] \\ T^2 \arrow[r, "\mu"] & T \end{tikzcd}$$

Definition 14 (Adjunction)

Given two categories $\mathcal{C}$ and $\mathcal{D}$, a functor $F : \mathcal{C} \to \mathcal{D}$ is a *left adjoint* to a functor $G : \mathcal{D} \to \mathcal{C}$ (and G is a *right adjoint* to F) if there is a natural isomorphism:

$$\mathrm{Hom}_{\mathcal{D}}(Fa, b) \cong \mathrm{Hom}_{\mathcal{C}}(a, Gb)$$

for all objects a in $\mathcal{C}$ and b in $\mathcal{D}$. This means that for every morphism $f : Fa \to b$ in $\mathcal{D}$, there is a unique morphism $g : a \to Gb$ in $\mathcal{C}$ such that the following diagram commutes:

$$\begin{tikzcd} Fa \arrow[r, "f"] & b \\ a \arrow[u, "F"'] \arrow[r, "g"'] & Gb \arrow[u, "G"'] \end{tikzcd}$$

We often denote this adjunction by $F \dashv G$.

Example (Monad from an Adjunction)

Given an adjunction $F \dashv G$ between categories $\mathcal{C}$ and $\mathcal{D}$, we can construct a monad on $\mathcal{C}$ as follows:

- The functor $T : \mathcal{C} \to \mathcal{C}$ is defined by $T = GF$.
- The unit $\eta : \mathrm{Id}_{\mathcal{C}} \to T$ is given by the natural transformation that assigns to each object a in $\mathcal{C}$ the morphism $\eta_a : a \to GFa$ corresponding to the identity morphism on Fa under the

adjunction.

- The multiplication $\mu : T^2 \to T$ is given by the natural transformation that assigns to each object a in $\mathcal{C}$ the morphism $\mu_a : GFGFa \to GFa$ corresponding to the composition of morphisms in $\mathcal{D}$ under the adjunction.

1.3 2-categories

Definition 15 (2-category)

A *2-category* C consists of:

- Cellular structure:
  - A collection C_0 of 0-cells $a, b, c, \dots$ (objects),
  - For each pair of 0-cells a, b, a collection $C_1(a, b)$ of 1-cells (morphisms) from a to b,
  - For each pair of 1-cells $f, g \in C_1(a, b)$, a collection $C_2(f, g)$ of 2-cells (2-morphisms) from f to g
- Identities:
  - A 1-cell $\mathrm{id}_a : a \to a$ for each 0-cell $a \in C_0$ (identity 1-cell),
  - A 2-cell $\mathrm{id}_f : f \Rightarrow f$ for each 1-cell $f \in C_1(a, b)$ (identity 2-cell)
- Composition:
  - A composition operation on 1-cells $g \circ f$ for every pair of composable 1-cells $f : a \to b$ and $g : b \to c$,
  - A vertical composition operation on 2-cells $\beta \circ \alpha : f \Rightarrow h$ for every pair of composable 2-cells $\alpha : f \Rightarrow g$ and $\beta : g \Rightarrow h$,
  - A horizontal composition operation on 2-cells $\beta \bullet \alpha : g \circ f \Rightarrow g' \circ f'$ for every pair of composable 2-cells $\alpha : f \Rightarrow f'$ and $\beta : g \Rightarrow g'$

  such that

  - 1-cell compositions are associative and identities are neutral
  - 2-cell vertical compositions are associative and identities are neutral
  - 2-cell horizontal compositions are associative and identities are neutral, and composition of 2-cell identities preserves 1-cell composition
  - the interchange law holds: for any composable 2-cells $\alpha, \alpha', \beta, \beta'$, we have

$$(\beta' \circ \beta) \bullet (\alpha' \circ \alpha) = (\beta' \bullet \alpha') \circ (\beta \bullet \alpha)$$

Example (2-category of Categories)

The 2-category **Cat** has:

- 0-cells: categories,
- 1-cells: functors between categories,
- 2-cells: natural transformations between functors.
- pointwise composition of functors as 1-cell composition,

- pointwise composition of natural transformations as vertical composition,
- horizontal composition of natural transformations defined by the composition of functors.

Definition 16 (Adjunction in a 2-category)
Given a 2-category C, an *adjunction* between 0-cells a and b in C consists of:

- Two 1-cells $f : a \to b$ and $g : b \to a$,
- Two 2-cells $\eta : \mathrm{id}_a \Rightarrow g \circ f$ (unit) and $\varepsilon : f \circ g \Rightarrow \mathrm{id}_b$ (counit),

such that the following triangle identities hold:

$$(f \bullet \eta) \circ (\varepsilon \bullet f) = \mathrm{id}_f \qquad (\eta \bullet g) \circ (g \bullet \varepsilon) = \mathrm{id}_g$$

1.4 Multicategories

Definition 17 (Multicategory)
A multicategory $\mathcal{U}$ consists of:

- A class of objects $\mathrm{Ob}(\mathcal{U})$,
- For each finite (possibly empty) sequence of objects $(a_1, a_2, ..., a_n)$ and an object b, a set of morphisms $\mathcal{U}(a_1, a_2, ..., a_n; b)$,
- For each object a, an identity morphism $\mathrm{id}_a : (a) \to a$,
- A composition operation that allows for the composition of morphisms in a way that generalizes the composition in a category, satisfying appropriate associativity and identity axioms.

We denote by **Multicat** the set of all multicategories.

Example (Category with Finite Products as a Multicategory)
Given a category $\mathcal{C}$ with finite products, we can construct a multicategory $\mathcal{U}$ where:

- The objects of $\mathcal{U}$ are the same as the objects of $\mathcal{C}$,
- The morphisms $\mathcal{U}(a_1, a_2, ..., a_n; b)$ are given by the morphisms in $\mathcal{C}$ from the product $a_1 \times a_2 \times ... \times a_n$ to b,
- The identities rely on the canonical isomorphism

$$\mathrm{id}_a := \prod_{i=1}^{n} a \cong a$$

- The composition is defined using the universal property of products and the composition in $\mathcal{C}$.

Example (Monoidal Category as a Multicategory)
A monoidal category $(\mathcal{C}, \otimes, I)$ can be viewed as a multicategory where:

- The objects of the multicategory are the same as the objects of $\mathcal{C}$,

- The morphisms $\mathcal{U}(a_1, a_2, ..., a_n; b)$ are given by the morphisms in $\mathcal{C}$ from $a_1 \otimes a_2 \otimes ... \otimes a_n$ to b (by nesting the monoidal product)

Definition 18 (Multifunctor)
A *multi-functor* $F : \mathcal{U} \to \mathcal{V}$ between multicategories $\mathcal{U}$ and $\mathcal{V}$ consists of two mappings:

- A mapping $F : \mathrm{Ob}(\mathcal{U}) \to \mathrm{Ob}(\mathcal{V})$ that assigns to each object a in $\mathcal{U}$ an object Fa in $\mathcal{V}$,
- A mapping $F : \mathcal{U}(a_1, a_2, ..., a_n; b) \to \mathcal{V}(Fa_1, Fa_2, ..., Fa_n; Fb)$ that assigns to each morphism $f : (a_1, a_2, ..., a_n) \to b$ in $\mathcal{U}$ a morphism $Ff : (Fa_1, Fa_2, ..., Fa_n) \to Fb$ in $\mathcal{V}$,

such that F preserves:

- Identities: For each object a in $\mathcal{U}$, $F\mathrm{id}_a = \mathrm{id}_{Fa}$,
- Composition: For morphisms $f : (a_1, a_2, ..., a_n) \to b$ and $g_i : (a_{i1}, a_{i2}, ..., a_{ik_i}) \to a_i$ in $\mathcal{U}$, we have
$$F(f \circ (g_1, g_2, ..., g_n)) = Ff \circ (Fg_1, Fg_2, ..., Fg_n).$$

Definition 19 (Multi-natural transformation)
Given two multi-functors $F, G : \mathcal{U} \to \mathcal{V}$ between multicategories $\mathcal{U}$ and $\mathcal{V}$, a *multi-natural transformation* $\eta : F \to G$ consists of a family of morphisms $\{\eta_a : Fa \to Ga\}_{a \in \mathrm{Ob}(\mathcal{U})}$ such that for every morphism $f : (a_1, a_2, ..., a_n) \to b$ in $\mathcal{U}$, the following diagram commutes:

$$\begin{tikzcd}
(Fa_1, Fa_2, ..., Fa_n) \arrow[r, "Ff"] \arrow[d, "(\eta_{a_1}, \eta_{a_2}, ..., \eta_{a_n})"'] & Fb \arrow[d, "\eta_b"] \\
(Ga_1, Ga_2, ..., Ga_n) \arrow[r, "Gf"'] & Gb
\end{tikzcd}$$

Definition 20 (Model)
Let $\mathcal{O}$ be an operad, and $\mathcal{U}$ be a multicategory. A *model* of $\mathcal{O}$ in $\mathcal{U}$ is a multi-functor $\mathcal{A} : \mathcal{O} \to \mathcal{U}$ that preserves the operadic structure, i.e., it respects the composition and identity of the operad.

Definition 21
Homomorphism of Models A *homomorphism* $h : \mathcal{A} \to \mathcal{B}$ between $\mathcal{O}$-models in $\mathcal{U}$ consists of a morphism $\underline{h} : (\underline{\mathcal{A}}) \to \underline{\mathcal{B}}$ such that for all $f \in \mathcal{O}_n$,

$$\underline{h} \circ (\mathcal{A}[[f]]) = \mathcal{B}[[f]](\underline{h}, \ldots, \underline{h})$$

Example (Category of Models)
Given an operad $\mathcal{O}$ and a multicategory $\mathcal{U}$, we can construct a category $\mathbf{Mod}(\mathcal{O},\mathcal{U})$ where:

- The objects are the models of $\mathcal{O}$ in $\mathcal{U}$,
- The morphisms are the homomorphisms between these models.

2 Some free algebras

Definition 22 (Finitary signature)

A *finitary signature* $\Sigma = (\mathrm{op}\,\Sigma, \mathrm{arity})$ consists of a set $\mathrm{op}\,\Sigma$ of *operators*, and an assigment $\mathrm{arity} : \mathrm{op}\,\Sigma \to \mathbb{N}$ associating to each operator in $\mathrm{op}\,\Sigma$ its *arity*.

Definition 23 (Algebra)

An *algrebra* $A = (\underline{A}, A[\![-]\!])$ over a finitary signature Σ consists of:

- A set $\underline{A}$ called the *carrier* of A,
- For each operator $f \in \mathrm{op}\,\Sigma$ of arity n, a function $A[\![f]\!] : \underline{A}^n \to \underline{A}$ called the *interpretation* of f in A.

Definition 24 (Terms)

Given a set X of variables and a finitary signature Σ, the set of *terms* $\mathrm{Term}_\Sigma(X)$ is defined inductively as follows:

- Every variable $x \in X$ is a term,
- If $f \in \mathrm{op}\,\Sigma$ is an operator of arity n and $t_1, t_2, \ldots, t_n$ are terms, then $f(t_1, t_2, \ldots, t_n)$ is a term.

Remark : The definition of algrebraic theory seen before can be reformulated as a finitary signature together with a set of equations between terms built from the operators in the signature.

Definition 25 (Presentation)

A *presentation* $\mathcal{T} = (\Sigma_\mathcal{T}, \mathcal{T}_a x)$ consists of a finitary signature $\Sigma_\mathcal{T}$ and a set $\mathcal{T}_a x$ of *axioms*, which are equations between terms in $\mathrm{Term}_{\Sigma_\mathcal{T}}(X)$ for some set of variables X.

A $\mathcal{T}$-model A is a $\Sigma_\mathcal{T}$-algebra that satisfies the axioms in $\mathcal{T}_a x$.

A Daily recap

- 05/05/2026: first day, talk with Ohad, building tour, admin stuff, setting up the git repo and starting to read the notes of Ohad
- 06/05/2026: reading Ohad's notes, starting to write my own notes. lots of definitions to try and understand how things work.

  TODO: dig in the paper about frexes to learn more about that (both frexes and frals) and write the relevant definitions down + better understand exactly what I have to do about the combinations of algebraic structures + look at what was done with involution to take inspiration